

Deadlines

You are asked to work in groups of 3. You will receive your group member notifications by the end of Week 2. Contact me (arnau.rios@fqa.ub.edu) if you foresee any issues.

Essay on a Landmark Paper

Your completed essay should be submitted by

10pm, Sunday 12th October 2025.

Each group should submit one single essay. The submission is *electronic*, via the [Virtual Campus](#) only. Please make sure your assignment is uploaded as a *single* Word or PDF file by the deadline. Note that only *one upload* is allowed, so make sure you only submit your very final version.

Presentation

The mini-conference will take place on the standard timetabled 2h session starting at

10am, Tuesday 21st October 2025.

Please **upload** the talk at the [Virtual Campus](#) for marking purposes (this can be done right after the presentation).

Peer assessment

The forms for peer assessment are available in the Coursework folder at the [Virtual Campus](#). You should download the form and mark the work of your 2 peers, following the template. Keep in mind that all marks need to be justified in writing. Save the document and upload it into the Coursework folder. The deadline for upload is:

10pm, Sunday 26th October 2025.

The submission is *electronic*, via the [Virtual Campus](#) only. The mark for each student will be an average of the mark coming from their peers.

Coursework submitted late will be subject to penalty. Your report will be run through **plagiarism** software!

Coursework details

The assessed coursework has 3 components:

1. A 4-page letter **essay** on a landmark paper of quantum mechanics. You should put the landmark paper in historical context, and describe its importance at the time and after it was published. You should also show evidence of having reproduced the arguments of the paper, either with numerical tools or updated figures. Your essay should be written in scientific style, using references to back up evidence and should include figures. This is **40 %** of the total coursework mark and the marking scheme will be available on the [Virtual Campus](#).
2. A 10-minute **presentation** (8 minute talk followed by 2 minutes of questions) during a mini-conference organised in a standard course session. This will amount **40 %** towards your coursework mark. The marking scheme is also available in the [Virtual Campus](#).
3. A **peer assessment** to be submitted according to a template on the [Virtual Campus](#). You will mark the contribution of each peer to the coursework components following scheme template. You are expected to work in groups of **3**, so each of you will have to submit a form for 2 of your peers. This will count **20 %** towards the final coursework mark.

All marking schemes are or will be available on the Coursework folder at the [Virtual Campus](#).

Essay

The essay should have a maximum of 4 pages (approximately **3500** words), excluding references. It should be written in a scientific style. Format is free, but the use of the LaTeX template uploaded in the [Virtual Campus](#) is recommended. The essay would normally follow the standard structure including the following sections:

1. An **introduction**, describing the historical setting of the landmark paper at the time
2. **Description** of the paper and its main results
3. **Discussion** of the relevance of the paper and evidence of having **reproduced** some of its content on your own
4. **Bibliography**

Keep in mind you need to justify and argue your ideas, in a way that gives a coherent theme running through the paper. Back up your information with scientific, academic references (avoid Wikipedia).

You will be marked on Command of the Subject (35 %), Argumentation & Language (35 %), Format (15 %) and Referencing (15 %). Pay particular attention in communicating your ideas well, but also do describe the scientific context of the time when the paper was published.

Your completed coursework should be submitted by **Sunday 12th October 2025**. The submission is electronic via the [Virtual Campus](#), where you will also find the marking scheme. If you have any questions about the assignment, please do let me know by email or speak to me during class.

Presentation

We will run a mock research mini-conference on the session of Week 4 on **Tuesday 21st October 2025**. You will be expected to give a 10 minute presentation (8 minute talk followed by 2 minutes of questions) describing the landmark paper and the conclusions of your essay. All students will attend the conference. Everyone is expected to ask questions - just as you would do in a standard research conference!

The presentation can either be given by one single appointed member of the group, or as a shared presentation. Keep in mind you should address the same points as the research paper: a good justification of the context of the landmark paper and how you have chosen to reproduce it.

The Marking Sheet for the presentation is available on the [Virtual Campus](#). The quality of the slide content, as well as the quality of the delivery will be marked.

Peer assessment

The peer group assessment is designed to provide some weight to individual contributions within a group. Additional marks will be provided to students that have contributed positively within the group according to their peers. Conversely, students that contribute below peer's expected averages, will have lower marks in this component.

One of the advantages of working as a member of a team is that you can all benefit from each other's strengths. The purpose of this exercise is to give recognition to the contributions that individuals make to the working of a group. In this peer marking exercise, you should:

- Read the example peer marking file, to understand how the process works.
- Fill the tables in the form **individually** to assess the relative contribution of each group member in each category.
- Provide a **justification** in the comment box on the form, using the evaluation criteria provided. Forms without specific written comments will not be accepted.

Free riding is to be **avoided** whenever possible. I encourage you to contact me via email (arnau.rios@fqa.ub.edu) if any issues arise around the group assessment. If you are a group member and suspect that another member is not contributing at the expected level, let me know and we will engage in an open discussion. Students that do not contribute will not only get a low peer assessment mark, but risk affecting their peer's marks on the paper and the presentation.

Your peer's final mark should be the sum of 5 different categories and counts 20% of the final coursework mark. The mark for each student will be an average of the mark coming from his/her two other peers.

You should upload to the [Virtual Campus](#) a single document containing the assessment of your 2 group members by **Sunday 26th October 2025**.